

Thermodynamic State Vector Non-Negative Entropy Assertion in Biospheric Simulation Engines

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Sprint 35 Report

Repository: <https://github.com/pascalranoroarijaona/WebOfLife>

Abstract

Simulating complex biophysical ecosystems and EarthPods requires strict adherence to physical conservation laws, most notably the First and Second Laws of Thermodynamics. In Sprint 35, we introduce the Thermodynamic State Vector Non-Negative Entropy Assertion module (`src/thermodynamics/state_validator.ts`). This utility provides robust runtime validation and functional monad pipelines enforcing absolute non-negativity constraints on system entropy ($S \geq 0$) and entropy generation rates ($\sigma = \frac{dS_{\text{gen}}}{dt} \geq 0$). By embedding these checks directly into thermodynamic state transitions, the Web of Life engine prevents unphysical backflow of thermodynamic time and guarantees compliance with the dissipation constraints of nonequilibrium thermodynamics.

1 Introduction and Thermodynamic Foundation

The Web of Life simulation architecture models EarthPods as open thermodynamic nonequilibrium systems where matter (carbon, nitrogen, phosphorus, water) and energy flow continuously. To maintain physical realism across biomass turnover, biogeochemical cycles, and energy dissipation, state transformations must satisfy fundamental physical boundaries:

1. **First Law (Mass & Energy Conservation):** Total atomic stock bounds and internal energy transformations remain strictly conserved across global biogeochemical pools.
2. **Second Law (Dissipation & Entropy Generation):** The total entropy change of the universe for any real, irreversible process must be non-negative:

$$\Delta S_{\text{univ}} \geq 0 \quad \text{and} \quad \sigma(t) = \frac{dS_{\text{gen}}}{dt} \geq 0 \quad (1)$$

Without programmatic enforcement, floating-point arithmetic drift or unconstrained phenomenological feedback loops can cause simulated systems to enter unphysical states characterized by negative absolute entropy or negative entropy production.

2 Thermodynamic Stock Transfer & Delta Equations

Let a thermodynamic state vector $\Gamma(t)$ at time t be defined as:

$$\Gamma(t) = \{U(t), T(t), S(t), \sigma(t), \vec{M}(t)\} \quad (2)$$

Where:

- $U(t)$ = Internal Energy (J)

- $T(t)$ = Absolute Temperature (K, $T > 0$)
- $S(t)$ = System Entropy ($\text{J} \cdot \text{K}^{-1}$, $S \geq 0$)
- $\sigma(t)$ = Entropy Generation Rate ($\text{J} \cdot \text{K}^{-1} \cdot \text{s}^{-1}$, $\sigma = \frac{dS_{\text{gen}}}{dt} \geq 0$)
- $\vec{M}(t)$ = Conservation mass stock vector (Carbon, Water, Nitrogen, Phosphorus)

2.1 Governing Equations

$$S(t) \geq 0 \quad \forall t \quad (3)$$

$$\sigma(t) = \frac{dS_{\text{gen}}}{dt} = \frac{d}{dt} \left(S_{\text{sys}} - \int \frac{dQ}{T} \right) \geq 0 \quad (4)$$

$$\sum_i M_i(t + \Delta t) = \sum_i M_i(t) \quad (5)$$

3 Architecture and Monad Implementation

Building incrementally upon existing thermodynamic structures, **StateValidator** provides static assertion hooks and validation result monads. The execution pipeline intercepts state transitions via functional monadic patterns (**neverthrow**), ensuring that any state violating entropy bounds triggers an immediate, traceable exception prior to EarthPod state commitment.

```
export class StateValidator {
  public static isValidEntropy(state: ThermodynamicState): boolean {
    return (
      Number.isFinite(state.entropy) &&
      state.entropy >= 0 &&
      Number.isFinite(state.entropyGenerationRate) &&
      state.entropyGenerationRate >= 0 &&
      state.temperature > 0
    );
  }

  public static assertNonNegativeEntropy(state: ThermodynamicState) {
    if (!StateValidator.isValidEntropy(state)) {
      return err(new ThermodynamicEntropyViolationError(state,
        'Second Law Violation: Entropy S=${state.entropy} >= 0 and sigma=${state.entropyGenerationRate} >= 0'));
    }
    return ok(state);
  }
}
```

4 Conclusion

Sprint 35 establishes a rigorous mathematical and software engineering foundation for thermodynamic validation within the Web of Life ecosystem. By enforcing Second Law constraints programmatically, we ensure that all simulated ecological and industrial processes remain anchored in physical reality.

Official Repository: <https://github.com/pascalranoroarijaona/WebOfLife>